EXPERIMENT NO: 2

TITLE: Design and Analysis of Industrial Power Supply Systems: Rectifiers, Inverters, and Choppers (Design can be implemented by using circuit design software, Simulink model or hardware

AIM: To design and analyze industrial power supply systems such as rectifiers, inverters, and choppers using MATLAB/Simulink.

APPARATUS:

Sr.No.	Equipment	Specification	Version
1	MATLAB IDE	Matlab Simulink model for circuit analysis	R2024a, R2025a

THEORY:

Industrial power supply systems are essential for converting electrical energy from one form to another to meet the requirements of various industrial applications. Power electronic converters such as rectifiers, inverters, and choppers form the core of these systems. They are used to regulate voltage, control current, and ensure efficient power delivery to loads.

Rectifiers→

Theory: Rectifiers convert AC voltage to DC voltage. They are categorized based on their control mechanism and number of phases (e.g., single-phase, three-phase, uncontrolled, controlled).

- **Uncontrolled Rectifiers:**

Utilize diodes to achieve rectification. The output DC voltage is fixed for a given AC input.

- **Controlled Rectifiers:**

Employ thyristors (SCRs) or other controllable switches, allowing for regulation of the output DC voltage by controlling the firing angle of the switches.

- **Full-wave vs. Half-wave:**

Full-wave rectifiers utilize both positive and negative halves of the AC cycle, resulting in a higher average DC output and reduced ripple compared to half-wave rectifiers.

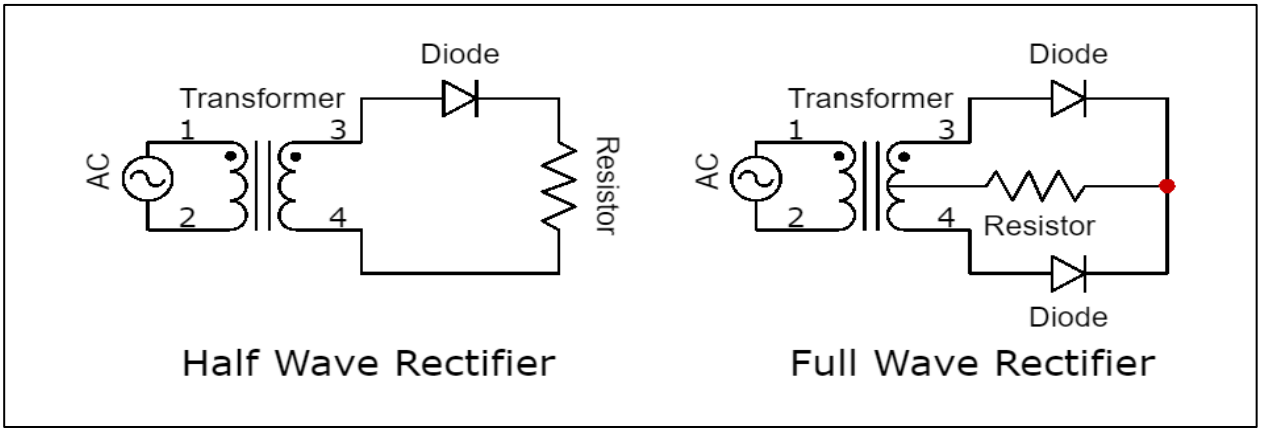
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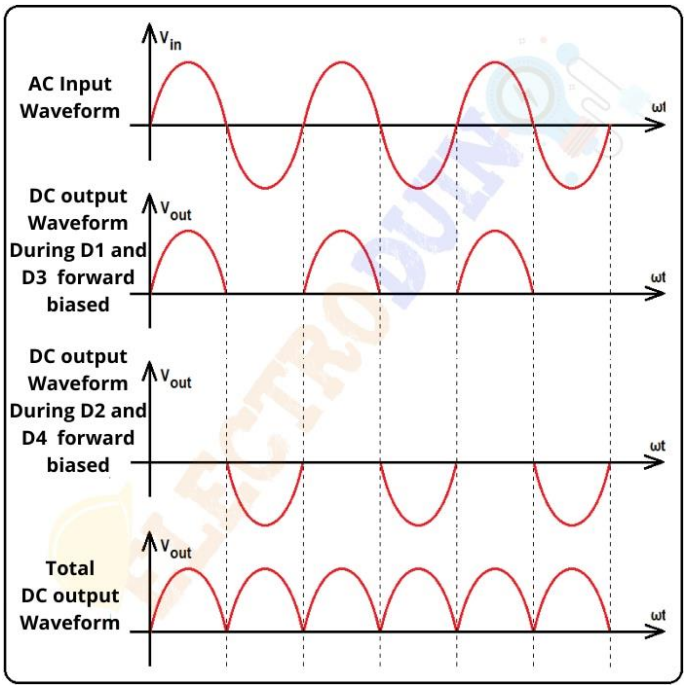
Inductors and capacitors are used in conjunction with rectifiers to smooth the DC output and reduce ripple, creating a more stable DC supply.

Simulink Implementation:

- Diodes, SCRs, AC voltage sources, and resistive/inductive loads are modelled using components from the Simscape Electrical library.
- Control signals for SCRs are generated using pulse generators or other control blocks.

- Scopes are used to visualize input and output waveforms, and powergui block is used for simulation settings.





Inverters→

Theory: Inverters convert DC voltage to AC voltage, often with variable frequency and voltage. They are crucial for applications like motor drives and uninterruptible power supplies (UPS).

- **Voltage Source Inverters (VSIs):**

The most common type, where the DC input acts as a voltage source, and the AC output voltage is controlled by switching devices.

- **Current Source Inverters (CSIs):**

The DC input acts as a current source, and the AC output current is controlled.

- **PWM (Pulse Width Modulation):**

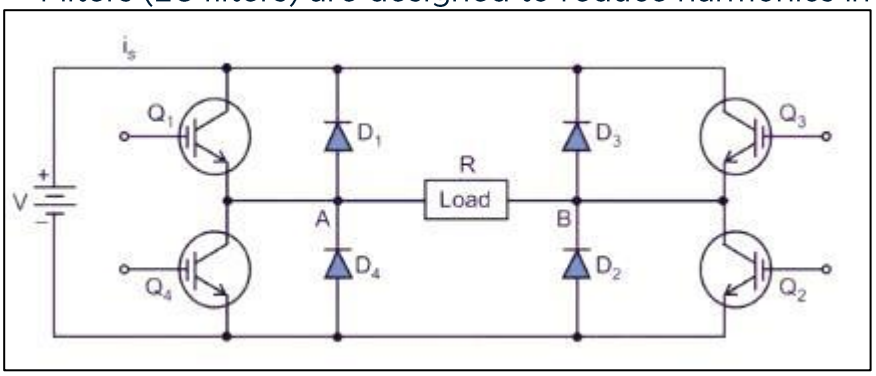
A common technique used to control the output voltage and frequency of inverters by varying the width of the pulses applied to the switching devices.

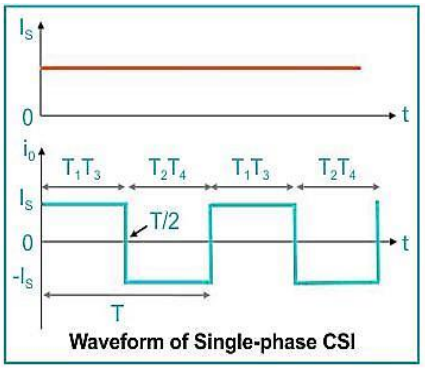
- **Harmonics:**

Inverter output waveforms typically contain harmonics, which need to be filtered to achieve a clean sinusoidal output.

Simulink Implementation:

- DC voltage sources, IGBTs or MOSFETs as switching devices, and inductive/resistive loads are used.
- PWM generators are implemented using control blocks to create the switching signals.
- Filters (LC filters) are designed to reduce harmonics in the output.





Choppers (DC-DC Converters)→

Choppers convert a fixed DC voltage to a variable DC voltage. They are widely used in applications requiring DC motor speed control, battery charging, and regulated DC power supplies.

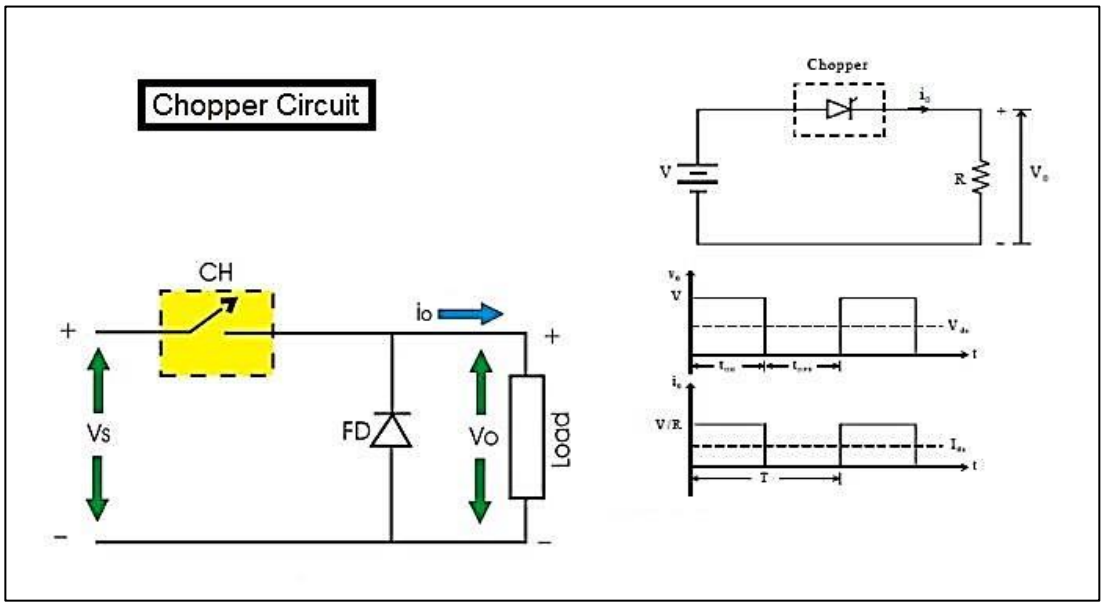
- **Buck Converter:** Steps down the input DC voltage.
- **Boost Converter:** Steps up the input DC voltage.
- **Buck-Boost Converter:** Can either step up or step down the input DC voltage.
- **Switching Frequency and Duty Cycle:** The output voltage of a chopper is controlled by varying the duty cycle (ratio of ON time to total switching period) of the switching device.
- **Continuous Conduction Mode (CCM) and Discontinuous Conduction Mode (DCM):** These modes describe the inductor current behaviour during operation and influence the design and performance.

Simulink Implementation:

- DC voltage sources, IGBTs or MOSFETs, diodes, inductors, and capacitors are used to build the converter topology.
- Pulse generators are used to control the switching device's duty cycle.
- Scopes are used to analyse input/output voltages and currents, and observe the ripple.

General Analysis in Simulink:

- **Steady-State Analysis:** Observing the system behaviour after transients have settled.
- **Transient Analysis:** Investigating the system's response to sudden changes in input or load.
- **Performance Evaluation:** Measuring parameters like efficiency, ripple factor, harmonic distortion, and power factor.
- **Control System Design:** Implementing feedback control loops to regulate output voltage or current.



RESULT:

Put the screenshot of the result of the experiment performed.(Each result page should contain Name, Roll No & Batch Name)

CONCLUSION:

Should be handwritten